



Pharm*Achieve*

HIV AND OPPORTUNISTIC INFECTIONS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

DEFINITIONS

HIV (Human Immunodeficiency Virus): RNA retrovirus

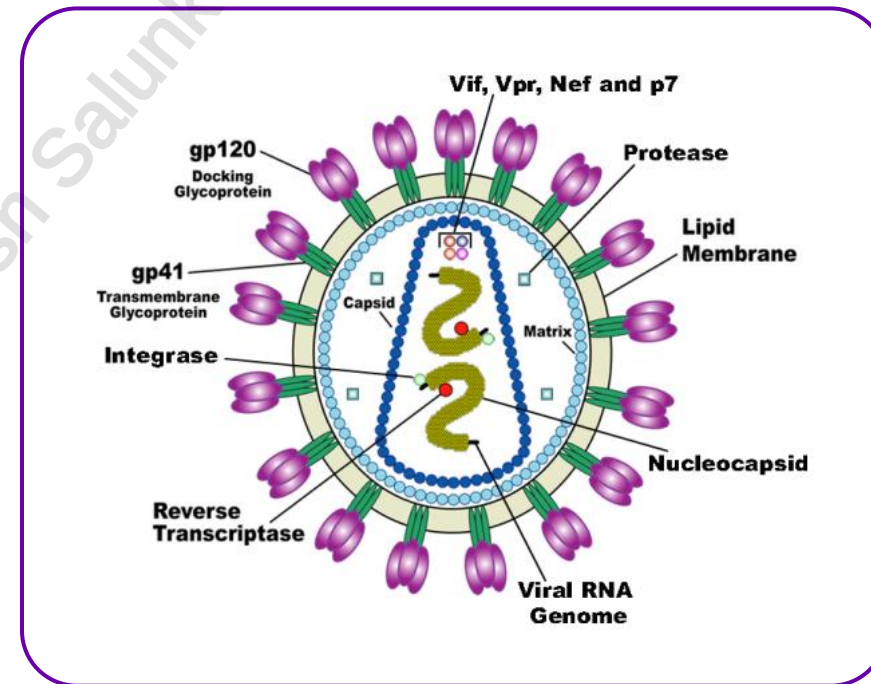
Two types of **RNA retrovirus**:

HIV-1:

- North America and Western Europe (focus of presentation)

HIV-2:

- Primarily in West Africa
- Less virulent
- Less easily transmitted than HIV-1



HIV TRANSMISSION

Sexually

- Receptive anal and vaginal intercourse are the most common routes of transmission
- HIV virus can be found in semen and cervical secretions

Parenterally

- Contaminated blood exposure such as needle sticks, IV injection with used needles (most common), receipt of contaminated blood products or organ transplants (1% of cases)

Perinatally

- “Vertical transmission” is the most common cause of pediatric HIV infections
- Most infections occur during or near the time of birth, thus treatment of the infected mother is important

Breastfeeding

- Can transmit HIV
- Formula feeding recommended

PATHOGENESIS

Mucosal exposure
to HIV virus

- Spread of infection to activated CD4⁺ lymphocytes
- HIV replicates in CD4⁺ cells and its virions are released into the blood

Day 0

Day 0-2

Day 4-11

Day 11 & onwards

- Virus collected by dendritic cells & carried to lymph node
- Fusion of dendritic cell and CD4⁺ lymphocytes occurs
- Transport of virus to the lymph nodes

Virus spreads to other organs
including spleen, brain, GI,
lymph nodes

CLINICAL PRESENTATION

Clinical latent period

- **Asymptomatic**
- Replication of virus in lymphoid tissue with continued immune system response
- Slow decline in CD4⁺ lymphocytes (over years)
- 'Set-point' of plasma HIV RNA
- As CD4⁺ count and viral load change, so does the risk of progression

Advanced Infection

- **Compromised immune system**
- Development of opportunistic infections and malignancies
- Lab abnormalities
 - Lymphopenia, leukopenia, anemia, thrombocytopenia
 - Increased viral load

GOALS OF THERAPY

- ✓ Restore CD4 count
- ✓ Slow progression of disease
- ✓ Improve quality of life
- ✓ Suppress viral replication
- ✓ Prevent transmission of HIV
- ✓ Prevent drug resistance
- ✓ Prevent/ treat opportunistic infections

CURRENTLY USED ART

NRTI	NNRTI	INSTIs	PI
abacavir (ABC) emtricitabine (FTC) lamivudine (3TC) tenofovir alafenamide (TAF) tenofovir disoproxil (TDF) zidovudine (ZDV)	doravirine (DOR) efavirenz (EFV) etravirine (ETV) nevirapine (NVP) rilpivirine (RPV)	bictegravir (BIC) cabotegravir (CAB) dolutegravir (DTG) elvitegravir/cobicistat (EVG/c) raltegravir (RAL)	Atazanavir (ATV) darunavir (DRV) Lopinavir (LPV) Tipranavir (TPV) Ritonaivr (RTV, r)



lenacapavir (capsid inhibitor) recently approved in Canada (November 2022) for patients with multiply resistant HIV-1

INITIATION OF THERAPY

PHARMACOLOGICAL MEASURES

General rules

- **Initiate ART** generally at diagnosis for most patients
- **Obtain a resistance test** (because of its more rapid turn-around time, a viral genotype is generally preferred over a viral phenotype)

Combination ART therapy

- **Initial HIV regimens include 2-3 HIV medicines from 2 or more drug classes**
- All guideline-recommended combination regimens are potent and can produce sustained virologic response
- **Currently, no evidence definitively recommends one type over another, and the selection is dependent on patient-specific factors and virologic factors such as virologic efficacy, toxicity, pill-burden, cost, dosing frequency, drug-drug interactions**

INITIATION OF THERAPY

PHARMACOLOGICAL MEASURES

Factors to consider when selecting initial regimen:

Comorbidity or conditions such as:

- Tuberculosis
- Hepatitis B
- Psychiatric disease
- Cardiovascular disease
- Chemical dependency
- Pregnancy or potential of pregnancy

Adherence potential and cost

Dosing convenience: pill burden, dosing frequency, food/fluid restrictions

Potential adverse effects

Potential drug interactions (DI) with other medications (consider ritonavir DI)

Result of genotypic drug resistance testing

Prior antiretroviral exposure

Renal function

INITIAL ANTIRETROVIRAL THERAPY

Recommended Regimen Options:

- BIC plus FTC/TAF
- DTG plus (TDF or TAF) plus (FTC or 3TC)
- DTG plus 3TC (if viral load < 500,000 copies/ml)

Note: Consider genotype and prior PrEP therapy



3TC = lamivudine
BIC = bictegravir
DTG = dolutegravir
FTC = emtricitabine

TAF = tenofovir
alafenamide
TDF = tenofovir
disoproxil fumarate

NNRTI-BASED COMBINATION THERAPY

1 NNRTI + 2 NRTI:

This regimen is commonly prescribed as initial therapy for treatment of naïve patients

Advantages:

- Lower pill burden compared with most PI regimens
- Allows the preservation of PI based therapies for future use

Disadvantages:

- Cannot be used in NNRTI resistant viral strains (since NNRTIs has low genetic barrier for development of resistance)

Efavirenz is the NNRTI of choice

Exceptions: Pregnancy, women planning to conceive or not using effective contraception

- May be associated with teratogenic effects in the first trimester of pregnancy
- In these cases, **nevirapine** may be used

INSTI-BASED COMBINATION THERAPY

1 integrase inhibitor + 2 NRTI:

This regimen is commonly prescribed as initial therapy for treatment of naïve patients

Advantages:

- Lower pill burden compared with most PI regimens
- Allows the preservation of PI based therapies for future use
- Can rapidly decrease viral load from baseline

Disadvantages:

- Drug-drug interactions
- GI intolerance

Considerations in choosing II will be dependent on patient population

- Newer single-pill regimens which include II (ie. BIC/TAF/FTC)
- Reduce the pill burden

PI-BASED COMBINATION THERAPY

1 PI + 1 PK Booster + 2 NRTI:

Each PI agent has its own unique characteristic based on:

- Clinical efficacy
- Adverse effects
- Pharmacokinetic properties and drug interactions

Adverse side effects:

Metabolic abnormalities: Dyslipidemia, fat maldistribution, insulin resistance

Ritonavir:

- Inhibits CYP 3A4, 2D6, 2C8 isoenzyme (induces others, see drug interactions)
- Addition of low-dose ritonavir to other PIs increases drug exposure and prolong plasma half lives of the active PI, this allows reduced dosing frequencies and pill burden



Disadvantage:

Increased risk of hyperlipidemia and drug interactions

POTENTIAL MODIFIERS OF INITIAL THERAPY

Clinical Issue	Modifier
CD4 < 200 cell/mm ³	Avoid rilpivirine regimens Avoid dolutegravir/ritonavir + raltegravir
Viral Load > 100,000 copies/mL	Avoid rilpivirine regimens Avoid abacavir/lamivudine+ (efavirenz or atazanavir/ritonavir) Avoid darunavir/ritonavir + raltegravir
Viral Load > 500,000 copies/mL	As above AND avoid dolutegravir + lamivudine
HLA-B*5701 + or unknown	Avoid abacavir
Unknown resistance profile starting ARV prior to genotyping	Avoid dolutegravir + lamivudine Avoid abacavir Use INSTI or PI based 3-drug regimen
Drugs that need to be taken with food	Atazanavir, darunavir, elvitegravir, rilpivirine
Drugs that need to be taken on empty stomach	Efavirenz
PrEP with INSTI	DRV/r/c + (TDF or TAF) + (3TC or FTC)

POTENTIAL MODIFIERS OF INITIAL THERAPY

Clinical Issue	Modifier
Kidney disease (CrCl < 60ml/min)	Avoid tenofovir DF Tenofovir AF ok if CrCl > 30 mL/min or ESRD on dialysis)
Osteoporosis	Avoid tenofovir DF
Psychiatric illness	Avoid efavirenz and rilpivirine
Dementia	Avoid efavirenz
Prolonged QT	Avoid efavirenz and rilpivirine
High Risk Cardiac Disease	Avoid PI or abacavir regimens if possible
Dyslipidemia	Avoid boosted PI, efavirenz, elvitegravir/cobicistat Tenofovir DF improves lipid levels, tenofovir AF does not
Poor adherence	Use PI or II based regimen (or if stable, consider transition to injectable LA)
Hepatitis B infection	Use tenofovir (AF or DF) + emtricitabine or lamivudine
HCV infection/TB	MONITOR FOR INTERACTIONS!!!!

INITIATION OF THERAPY

PHARMACOLOGICAL MEASURES

Co-infection with Hepatitis B (HBV) Virus:

When treatment for HBV co-infection is indicated:

- **A 2-NRTI combination + either a PI + PK enhancer or INSTI or NNRTI**
- **Dolutegravir/lamivudine is not recommended for HBV co-infection**
- Initiation of a fully suppressive antiretroviral regimen against HIV that includes **tenofovir** and either **lamivudine** or **emtricitabine** is recommended in order to prevent drug resistance
- If ART is not initiated, HBV therapy should include only agents with the least potential for selecting HIV resistance mutations

ADHERENCE

- Treatment failure and development of viral resistance is directly related degree of ARV adherence
- > 95% adherence to ART is ideal to:
 - ✓ Ensure maximal viral suppression, although resistance can occur despite taking 100% of doses
 - ✓ Preserve immune function

Predictors of poor adherence:

- Lack of a trusting relationship between patient and provider
- Active illicit drug and/or alcohol use
- Active mental illness
- Lack of patient education regarding medications
- Lack of reliable access to medical care



ADHERENCE

How to improve adherence:

- ✓ **Assess patient's readiness to start therapy**
- ✓ **Provide information on:**
 - Drugs & doses
 - Medication timing with respect to meals and daily routine
 - Short- and long-term toxicities (including measures to be taken if problems occur)
- ✓ **Provide adherence aids:**
 - Medication schedules, timers
 - Daily or weekly pill boxes
 - Language-appropriate written drug and disease information
- Once daily/monthly regimens may increase the likelihood of success by improving patient's adherence

PREGNANCY

Prevention of vertical transmission is possible/likely if using ART during pregnancy

- ✓ Focus on appropriate screening pre-conception and during pregnancy
- ✓ Perinatal use of ART (i.e. zidovudine) in infants born to HIV infected mothers
- ✓ Early initiation of maternal ART
 - Preferred regimens
 - (Raltegravir, dolutegravir) + (tenofovir/emtricitabine, tenofovir/lamivudine or abacavir/lamivudine)
 - Drugs to avoid
 - Efavirenz
 - Elvitegravir/cobicistat
 - Bictegravir

ART ADRS QUICK REFERENCE

ADR	NRTI	NNRTI	INSTIs	PI
Bone Mineral Density	TDF>TAF			
Cardiac Conduction Changes		RPV, EFV (QTc)		ATV/r, LPV/r (PR)
Cardiovascular disease	ABC			DRV, LPV/r
Dyslipidemia (Inc TG, LDL, HDL)	TAF	EFV	EVG/c	Boosted Pis
Dysglycemia	ZDV			LPV/r
GI ADRs	ZDV		EVG/c	All, LPV/r (diarrhea)
Hepatotoxicity	(if withdrawn in HBV)	EFV, NVP	DTG	All
Hypersensitivity reactions	ABC	NVP	RAL	
Neuropsychiatric ADRs		EFV>RPV>DOR, ETR	All	
Rash	FTC	All	All	ATV, DRV, LPV/r
SJS		NVP>EFV,ETR, RPV	RAL	DRV, LPV/r, ATV
Weight Gain			All	

ABC=abacavir; ATV=atazanavir; DOR=doravirine; DRV=darunavir; DTG=dolutegravir; EFV=efavirenz, ETR=etravirine, EVG/c=elvitegravir/cobicistat; FTC=emtricitabine; LPV/r=lopinavir/ritonavir; NVP=nevirapine; RAL=raltegravir; RPV=rilpivirine; TAF=tenofovir alafenamide; TDF=tenofovir disoproxil fumarate ZDV=zidovudine

ADR MANAGEMENT

ADR	SUGGESTED TREATMENT
Diarrhea	Loperamide, oral rehydration
Nausea	• dimenhydrinate • ondansetron
Rashes	May be mild and self-limited or may represent severe ADR therefore REFERRAL
GI upset	Take with food: • Except efavirenz
MI/CVD risk	Manage risk factors: • Hyperlipidemia (prefer for tenofovir DF if possible) • Hypertension • Insulin resistance • Modify lifestyle/diet

ADR MANAGEMENT

ADR	SUGGESTED TREATMENT
Diabetes	• Diet and exercise • Switch from PI to NRTI or INSTI • Start diabetic medications
Lipohypertrophy	Liposuction
Osteopenia/Osteoporosis	• Exercise • Calcium/vitamin D supplementation • Bisphosphonate therapy
Peripheral neuropathy	• Gabapentin/pregabalin, amitriptyline • If drug-induced (i.e. NRTI's), consider medication change
HIV induced nephropathy	• Start HIV medications early since this is due to untreated HIV causing inflammatory injuries

MANAGEMENT OF DRUG INTERACTIONS

Suggested approach:

- ✓ Obtain full patient medical history
- ✓ Familiarize yourself with characteristics of the drug (e.g. are they metabolized by CYP450s?)
- ✓ Consult references to confirm the drug interaction
- ✓ Evaluate the clinical significance of the interaction



Assess requirement for intervention:

- ☐ Do not intervene, just monitor (when interacting drug is only used for a short term)
- ☐ Separate administration times for absorption interactions
- ☐ Change dose for pharmacokinetic interactions
- ☐ Change the drug

DRUG-DRUG INTERACTIONS

METABOLISM AND ELIMINATION

DRUGS/ DRUG CLASS	METABOLIC MECHANISM	CO-ADMINISTERED DRUGS	INTERACTION EFFECT
All PIs	Inhibit CYP3A4	CYP3A4 substrates: - Calcium channel blockers - Statins	Co-administration may result in accumulation of the substrate drug and thus predispose patients to: - Hypotension - Bradycardia - Rhabdomyolysis

DRUG-DRUG INTERACTIONS

METABOLISM AND ELIMINATION

DRUG CLASS	METABOLIC MECHANISM	CO-ADMINISTERED DRUGS	INTERACTION EFFECT
PIs	Substrates and inhibitors of P-glycoprotein (Pgp) efflux transporter	Other Pgp substrates: • Digoxin • Amiodarone • Colchicine	While PIs may help increase entry of other PIs into sanctuary sites, their inhibition of Pgp may also lead to toxicity from other Pgp substrates

DRUG-DRUG INTERACTIONS

Protease Inhibitors

Drug	↑ gastric pH	P-GP	CYP substrate	CYP inhibitor	CYP inducer
Atazanavir	↓	Substrate, inhibitor	3A4	3A4, 2C8	No
Atazanavir/ Ritonavir	↓	Substrate, inhibitor	3A4, 2D6	3A4, 2D6, 2C8	1A2, 2B6 2C8, 2C9, 2C19
Atazanavir/ Cobicistat	↓	Substrate, inhibitor	3A4	3A4, 2D6, 2C8	No
Darunavir/ ritonavir		Substrate, inhibitor	3A4, 2D6	3A4, 2D6	1A2, 2B6 2C8, 2C9, 2C19
Darunavir/ cobicistat		Substrate, inhibitor	3A4	3A4, 2D6	No
Lopinavir/ ritonavir		Substrate	3A4, 2D6	3A4, 2D6	1A2, 2B6 2C8, 2C9, 2C19
Tipranavir/ ritonavir		Substrate, inducer	3A4, 2D6	3A4, 2D6	? (see ritonavir)

DRUG-DRUG INTERACTIONS

Integrase Strand Transfer Inhibitors

Drug	Chelation	P-GP	CYP substrate	CYP inhibitor	CYP inducer
Bictegravir	All absorption inhibited by polyvalent cations	substrate	3A4	No	No
Dolutegravir		substrate	3A4 (weak)	No	No
Elvitegravir/ cobicistat		substrate	3A4	3A4, 2D6	2C9
Raltegravir			No	No	No

DRUG-DRUG INTERACTIONS

NNRTIs

Drug	 gastric pH	CYP substrate	CYP inhibitor	CYP inducer
Doravirine		3A4, 3A5	No	No
Efavirenz		2B6, 2A6, 3A4	3A4	3A4, 2B6, 2C19
Etravirine		3A4, 2C9, 2C19	2C9, 2C19	3A4
Nevirapine		3A4, 2B6	No	3A4, 2B6
Rilpivirine		No	No	No

DRUG-DRUG INTERACTIONS

NRTIs

Drug	 gastric pH	P-GP	CYP substrate	CYP inhibitor	CYP inducer
Abacavir					
Emtricitabine					
Lamivudine					
Tenofovir DF		Substrate			
Tenofovir AF		Substrate			
Lopinavir/ ritonavir					
Zidovudine					

MONITORING PARAMETERS

INDICATION	HIV RNA/CD4	INFORMATION
Acute retroviral syndrome	Yes	Consider resistance testing
New HIV Diagnosis	Yes	Determine viral set-point
Initiation of ART or change in therapy	Test at initiation of ART	
	Every 3-4 months after start for first 2 years*	Treatment effectiveness
	Every 12 months after 2 years **	Durability of response

*Every 3-4 months during the first 2 years of ART or if the viremia develops while the patient is on ART or if the CD4 count <300 cells/mm³ or if ART is delayed.

**CD4 count every 12 months if after 2 years the CD4 count is consistently 300-500 cell/mm³ while the patient is on ART or if the CD4 count is >500 cells/mm³.

THERAPEUTIC TIPS ON HIV INFECTION

- ✓ Ensure patient adherence with ART regimens
- ✓ Support, de-stigmatize and encourage harm reduction and testing strategies
- ✓ Manage ADRs related to ART therapy including hyperglycemia, dyslipidemia, rash, neuropathy
- ✓ Ensure starting appropriate prophylaxis treatment therapy for antifungal infection, CMV, herpes simplex based on CD4 count
- ✓ Monitor for drug interactions
- ✓ Develop relationships!
- ✓ Monitor for drug-drug, drug-food interactions



UHN | Immunodeficiency Clinic



HIVinfo.NIH.gov



University of Liverpool

- ✓ Keep up-to-date with changing information regarding new drugs and regimens



CATIE



CLINICAL INFO HIV.gov

OPPORTUNISTIC INFECTIONS

Major cause of mortality and morbidity in HIV+ individuals



Generally, as CD4+ count decreases, susceptibility to infections increases



Some infections are independent of CD4+ count

OPPORTUNISTIC INFECTIONS- PROPHYLAXIS

- ✓ Update all routine vaccinations
 - No live vaccines if CD4 $\leq$ 200 cell/mL
 - Possible exception: MMR, Varicella
- ✓ If not immune, Hepatitis A vaccine
- ✓ If not immune, Hepatitis B vaccine
- ✓ Annual influenza vaccination with inactivated vaccine
 - Live vaccine contraindicated
- ✓ Pneumococcal vaccine

Consider prophylaxis for HPV, HSV, meningococcal disease

OPPORTUNISTIC INFECTIONS- PROPHYLAXIS

If positive purified protein derivative test (> 5mm) or interferon-gamma release assays support TB:

- Provide latent infection regimen for *Mycobacterium tuberculosis*
 - Preferred prophylaxis: rifampin daily x 4 months or rifapentine + INH weekly x 12 weeks (note: drug interactions OR
 - INH daily x 9 months + pyridoxine

If CD4 < 200 cells/ uL or thrush:

- Provide prophylaxis for *Pneumocystis jiroveci* pneumonia (PCP)
 - Preferred prophylaxis: sulfamethoxazole-trimethoprim
 - Alternate: atovaquone, pentamidine, dapsone

OPPORTUNISTIC INFECTIONS- PROPHYLAXIS

If CD4 < 100 cells/ uL and positive/unknown for *Toxoplasma gondii*:

- Provide prophylaxis for *T. gondii* encephalitis
 - Preferred prophylaxis: **sulfamethoxazole-trimethoprim**

If CD4 < 50 cells/ uL:

Provide prophylaxis for the following:

- *Mycobacterium avium* complex (MAC) [only if not on ART!]
 - **Macrolide (azithromycin once weekly or clarithromycin twice daily)**
- Recurrent thrush or history of esophageal candidiasis:
 - **fluconazole**

OPPORTUNISTIC INFECTIONS – TREATMENT (SELECT)

INFECTION	GENERAL TREATMENT
Bartonellosis (<i>Bartonella henselae</i> and <i>B. quintana</i>)	Preferred: doxycycline and consider combination therapy for CNS/cardiac disease
Candida Species – Oral thrush	Initial: Topical agents (e.g. nystatin suspension) (note: if esophageal, requires systemic therapy), consider prophylaxis
Candida Species – Vaginal candidiasis	First line: Topical azole antifungal creams or tablets/suppositories
<i>Cryptococcus neoformans</i>	<i>Induction therapy:</i> IV amphotericin B + flucytosine for 2–6 weeks, then oral or IV fluconazole for $\geq$ 8-weeks <i>Maintenance therapy:</i> Oral fluconazole daily
Cytomegalovirus (CMV)	<i>Induction therapy:</i> Oral valganciclovir, IV ganciclovir, or IV foscarnet X 14–21 days <i>Maintenance therapy:</i> Oral valganciclovir, IV foscarnet or IV cidofovir daily

OPPORTUNISTIC INFECTIONS - TREATMENT

INFECTION	GENERAL TREATMENT
Intestinal Protozoa	Only consistently effective therapy: Immune reconstitution with antiretroviral therapy
Mycobacterium avium Complex (MAC)	Multidrug regimens: Base drugs = oral macrolide (clarithromycin or azithromycin) + ethambutol; add-on drugs depends on circumstance (include rifabutin (or rifampin), ciprofloxacin or levofloxacin, or amikacin)
<i>Pneumocystis jirovecii</i>	Standard therapy: Oral or IV sulfamethoxazole/trimethoprim x 21 days
<i>Toxoplasma gondii</i>	Standard therapy: pyrimethamine + leucovorin + sulfadiazine x 4–8 weeks OR sulfamethoxazole/trimethoprim, followed by maintenance therapy

CASE 1

HH is a 42-year-old male who presents with a recent diagnosis of HIV. HH has an HIV+ partner and has recently tested positive for HIV on routine screening. He has noticed lethargy, myalgia, and oral ulcers in the last month. HH has a history of depression for which he takes escitalopram 20 mg PO daily. He has an allergy to penicillin resulting in a rash. He smokes 5-10 cigarettes per day, drinks 2 alcoholic drinks per day, and smokes cannabis recreationally.

Objective findings:

- CD4=380
- Viral load=450,000 copies/mL
- LFTs normal
- Hep B/C negative
- TB negative
- SCr=67



CASE 1 - QUESTIONS

1. Which of the following combinations would be a suitable ART regimen for HH?
 - a) Emtricitabine + tenofovir + raltegravir
 - b) Tenofovir + emtricitabine + dolutegravir
 - c) Efavirenz + abacavir + lamivudine
 - d) Abacavir + darunavir + indinavir
2. What would you counsel HH on?
 - a) Take with food
 - b) Take medication daily and do not miss doses
 - c) Can use dimenhydrinate if experience nausea
 - d) Follow up with MD in 2-4 weeks after starting therapy
3. HH returns in 3 months. He is noticeably thinner, and you note he has not filled his ART since starting therapy. His CD4 is now 190. What would you recommend to his MD?
 - a) Sulfamethoxazole/trimethoprim
 - b) Azithromycin
 - c) Obtain vaccinations
 - d) Dapsone
4. What would you suggest to help HH with adhering to ART?
 - a) Daily medication reminders/alerts
 - b) Single pill regimen
 - c) Ensure depression is adequately treated
 - d) Injectable regimen

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

HIV ART APPENDIX

Qualifying Exam Preparatory Course

NRTI

Mechanism of action:

- Competitively incorporate into a growing viral DNA strand and cause strand termination as they have no 3' end to which another nucleotide can be attached to
- NRTIs require intracellular tri-phosphorylation to become active
 - Increases intra-cellular half-life
 - Greater than plasma half-life



Examples of NRTI class:

NRTIs: zidovudine (AZT, Retrovir®), lamivudine (3TC)

abacavir (ABC, Ziagen®), tenofovir (Viread®), emtricitabine (FTC)

NRTI

Adverse effects common to all NRTIs:

- Nausea/vomiting
- Headache
- Dyslipidemia (TAF>TDF)
- Hepatic effects
- Peripheral neuropathy



Rare adverse effect: Lactic acidosis (much less common with newer agents)

Specific adverse effects:

tenofovir:

- Renal tubular dysfunction (TDF>TAF)
- Asthenia
- Diarrhea
- Flatulence
- Decreased BMD (TDF>TAF)

abacavir:

- Hypersensitivity reaction (need to complete HLA-B*5701 testing prior to starting to avoid rash)

zidovudine:

- Associated with anemia/bone marrow suppression
- Only available IV ARV

NNRTI

Mechanism of Action:

- Binds to a **hydrophobic region** of the **HIV-1 reverse transcriptase** that is **not the active site**
- It changes the enzyme conformation and **reduces the activity**



Examples of NNRTIs:

- efavirenz
- nevirapine
- etravirine (Intelence®)
- rilpivirine (Edurant®)
- doravirine (Pifeltro®)

Advantages:

- Less dyslipidemia and fat maldistribution than protease inhibitor-based regimens
- Protease inhibitors are preserved for future use

Concerns:

- Low barrier to resistance
- Cross resistance is common within the NNRTIs
- Resistance to one NNRTI may confer resistance to other NNRTIs
 - Single mutation (K103 mutation) can lead to resistance

NNRTI

Metabolism:

- Metabolized by hepatic CYP450, thus susceptible to many drug interactions and hepatotoxicity
- Nevirapine dose escalation is required after the initial 2 weeks due to auto-induction

Common adverse drug reactions:

- Nausea/vomiting
- Rash
- Drowsiness (especially with efavirenz)

Specific adverse effects:

nevirapine:

- Symptomatic hepatitis
- Rash (reason why dose is started as once daily)

efavirenz:

- CNS side effects including vivid dreams
- Teratogenic (do not start in childbearing women)
- Once-daily dosing (better adherence)

etravirine:

- Most effective when NNRTI mutations are minimal

PROTEASE INHIBITORS (PI)

Mechanism of Action:

- Bind to the **active site of the protease enzyme** to **prevent cleavage of HIV precursor proteins** to smaller, functional proteins required for **HIV maturation**



Metabolic Issues:

- Older agents when used in higher doses (ritonavir, saquinavir, indinavir), which led to fat maldistribution (facial wasting with fat belly)
- Dyslipidemia (increased TGs, LDLs) & hyperglycemia
- Elevated transaminases (increased aminotransferase activity)
- All are substrates of CYP3A4, p-GP:
 - Metabolism altered by inducers and inhibitors
 - Most inhibit CYP3A4 (exception: tipranavir 2D6 inhibitor)
 - Significant potential for drug-drug and drug-food interactions

PROTEASE INHIBITORS (PI)

Examples and adverse effects:

atazanavir (Reyataz®): Hyperbilirubinemia, 1st degree AV block

darunavir (Prezcobix®): GI intolerance, rash

lopinavir/ritonavir (Kaletra®): GI intolerance (more with once daily dosing), asthenia

tipranavir (Aptivus®): Hepatotoxicity, rash, GI intolerance, intracranial hemorrhage

ritonavir (RTV, Norvir®): GI intolerance; used as boosted drug rather than a PI



Boosted ritonavir doses can decrease the GI side effects related to high dose monotherapy PI, decrease # of tablets and frequency

No longer used: indinavir (Crixivan®), nelfinavir (Viracept®), saquinavir (Invirase®)

COBICISTAT

Indication:

- Booster for atazanavir, darunavir, or elvitegravir
- Recently emerged alternative to ritonavir

Dose: 150 mg oral tablet once daily in treatment naïve or experienced

- Administer with food
- Generally, part of a combination tablet

Adverse drug reactions:

- Most common are jaundice and rash
- **Generally well tolerated**, similar tolerability to ritonavir

Mechanism of action:

- **Potent CYP3A inhibitor**
- **Increases systemic exposure of CYP3A substrates** where bioavailability is limited and half-life is shortened due to CYP3A-dependent metabolism
- **No activity against HIV on its own**

Avoid: CYP3A inducers (efavirenz, nevirapine), CYP3A substrates other than atazanavir or darunavir

Advantages:

- Allows lower/less frequent dosing of ART
- Improves adverse reaction profile

INTEGRASE STRAND TRANSFER INHIBITORS (INSTIS)

Mechanism of action:

- **Prevents insertion** of HIV-1 DNA into host cell genome

Indications:

- Advanced disease/extensive prior therapy (due to clinical trials requirements initially studied in treatment-experienced patients)
- **Now approved in treatment-naïve patients**

raltegravir (Isentress™):

- Does not inhibit/induce CYP450 enzymes or P-glycoprotein
- Increased serum ALT, low incidence of lipid changes

Dose: 400 mg BID in treatment experienced patients, 800mg daily dose in treatment naïve patients

Tips and drug-drug interactions:

- Can be taken with or without food, separate administration with Mg^{2+}/Al^{3+} antacids
- Caution with unboosted atazanavir (inhibits UGT=↑raltegravir)
- Avoid rifampin

Adverse drug reactions common to all IIs:

- **Nausea**
- **Diarrhea**
- **Headache**
- **Rash**
- **Upper respiratory tract infection**
- **Fatigue**
- **Insomnia**
- **Possible depression**
- **Weight gain**

INTEGRASE STRAND TRANSFER INHIBITORS (INSTIS)

dolutegravir (Tivicay®)

Dose: 50mg once daily tablet in treatment naïve, BID in treatment experienced, take with food

- May have some activity against raltegravir and elvitegravir resistant viruses
- Increased serum ALT
- Hyperglycemia reported in 10-14%
- Separate administration from medications with polyvalent cations
- No need for pharmacokinetic boosting
- Less expensive than raltegravir

elvitegravir

Dose: 85 mg or 150 mg tablet once daily, take with food

- No longer available as single-drug tablet
- Approved in treatment naïve and certain treatment-experienced patients as **Stribild** and **Genvoya**
- Use in antiretroviral-experienced individuals is currently being investigated
- Separate antacid administration by at least 2 hours
- Requires boosting (usually by cobicistat) which increases drug interaction potential

INTEGRASE STRAND TRANSFER INHIBITORS (INSTIS)

bictegravir (in Biktarvy® with emtricitabine/tenofovir AF)

Dose: 50mg once daily tablet, may have some activity against raltegravir and elvitegravir resistant viruses

- Increases metformin concentration
- Avoid cations 2h before or 6h after food

cabotegrevir (rilpivirine + oral [Vocabria®] or injected with rilpivirine [Cabenuva®])

Dose:

ORAL LEAD-IN	I.M. INITIATION INJECTIONS	I.M. CONTINUATION INJECTIONS
Month 1	Month 2	Month 3 onwards
VOCABRIA® 30 mg cabotegrevir tablet once daily EDURANT® 25 mg rilpivirine tablet once daily	CABENUVA® 3 mL (600 mg) cabotegrevir injection and 3 mL (900 mg) rilpivirine injection	CABENUVA® 2 mL (400 mg) cabotegrevir injection once monthly and 2 mL (600 mg) rilpivirine injection once monthly

Adverse effects:

- Injection site reaction
- Elevated LFT, lipase
- Weight gain

Drug Interactions: due to CYP3A4 substrate

REFERENCES

1. Anderson PL, Kakuda TN, Fletcher CV. Chapter 103. Human Immunodeficiency Virus Infection. In: DiPiro JT, Talbert RL, Yee GC, Matzke GR, Wells BG, Posey L. eds. *Pharmacotherapy: A Pathophysiologic Approach*, 9e. New York, NY: McGraw-Hill; 2014.
2. Bondy L, Walmsley S. Opportunistic Infections in HIV-positive Patients. *RxTx*. 2014. Available at: <https://www.e-therapeutics.ca/print/new/documents/CHAPTER/en/c0102#c0102n00026>.
3. Bristol-Myers Squibb. Evotaz Full Monograph. *RxTx*. Available at: <https://www.e-therapeutics.ca>.
4. Centers for Disease Control and Prevention. "Guidelines for Prevention and Treatment of Opportunistic Infections in HIV-Infected Adults and Adolescents." MMWR 2009;58 (No. RR-4) Available at: http://www.aidsinfo.nih.gov/contentfiles/lvguidelines/adult_oi_041009.pdf.
5. Cobicistat. *Lexicomp*
6. Coffey S. Drug Interactions: Integrase Inhibitors and Cations. HIV InSite. <http://hivinsite.ucsf.edu/insite?page=hmq-1408-03>.
7. DiPiro, Joseph T. et al. "Human Immunodeficiency Virus Infection" *Pharmacotherapy: A Pathophysiologic Approach*. Ch. 13, 8th edition. The McGraw-Hill Companies, Inc. 2011.
8. Dolutegravir. In: Lexi-Drugs. Hudson, OH: Lexi-Comp, Inc.
9. Efavirenz. In: Lexi-Drugs. Hudson, OH: Lexi-Comp, Inc.
10. Hand E, Dzintars K. Chapter 35. HIV Infection and AIDS. In: Attridge RL, Miller ML, Moote R, Ryan L. eds. *Internal Medicine: A Guide to Clinical Therapeutics*. New York, NY: McGraw-Hill; 2013.
11. Harris M, Montessori V, Montaner J. HIV Infection. *RxTx*. 2014. Available at: <https://www.e-therapeutics.ca/print/new/documents/CHAPTER/en/c0101>.
12. Kreutzweizer Denise. HIV infection: part 2. Presented as part of Pharmacy 320. February 13, 2020; Waterloo, ON.
13. Marzolini C, Gibbons S, Khoo S, Back D. Cobicistat versus ritonavir boosting and differences in the drug–drug interaction profiles with co-medications. *J Antimicrob Chemother*. 2016;71 (7): 1755-1758. doi: 10.1093/jac/dkw032.
14. McNicholl IR, Coffey S. *Clinically Significant Drug Interactions with Dolutegravir*. University of California San Francisco. <http://hivinsite.ucsf.edu/insite?page=hmq-1311-12>.
15. Panel on Opportunistic Infections in HIV-Infected Adults and Adolescents. Guidelines for the prevention and treatment of opportunistic infections in HIV-infected adults and adolescents: recommendations from the Centers for Disease Control and Prevention, the National Institutes of Health, and the HIV Medicine Association of the Infectious Diseases Society of America. Available at http://aidsinfo.nih.gov/contentfiles/lvguidelines/adult_oi.pdf.
16. US Department of Health & Human Services. *Antiretroviral drugs used in the treatment of HIV infection*. <http://www.fda.gov/ForConsumers/byAudience/ForPatientAdvocates/HIVandAIDSactivities/ucm118915.htm>.
17. Panel on Antiretroviral Guidelines for Adults and Adolescents. Guidelines for the Use of Antiretroviral Agents in Adults and Adolescents with HIV. Department of Health and Human Services. Available at <https://clinicalinfo.hiv.gov/en/guidelines/hiv-clinical-guidelines-adult-and-adolescent-arv>

CHANGE LOG

July 2025

- Complete overhaul of content
- Removal of superfluous material
- Moved drug information to appendix
- Case updated

October 2025

- No content changes made

March 2026

- Slide 2: Updated NAPRA competencies